Arakelov Geometry

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2 On rings of integers

Let K be a number field with ring of integers $\mathcal{O}_K$.

2.1 Hermitian line bundles over rings of integers

Definition 2.1. A *Hermitian line bundle* $\overline{\mathcal{L}}$ over $\operatorname{Spec}\mathcal{O}_K$ is a pair $(\mathcal{L}, (\|\cdot\|_\sigma)_{\sigma:K\to\mathbb{C}})$ where

- $\mathcal{L}$ is a line bundle over $\operatorname{Spec}\mathcal{O}_K$, i.e., a finitely generated projective $\mathcal{O}_K$-module of rank 1.
- For each embedding $\sigma : K \to \mathbb{C}$, $\|\cdot\|_\sigma$ is a hermitian metric on the complex vector space $\mathcal{L}_\sigma := \mathcal{L} \otimes_{\mathcal{O}_K,\sigma} \mathbb{C}$.

Example 2.2. Let $K = \mathbb{Q}(\sqrt{-5})$ with ring of integers $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$. The class number of K is 2, and the ideal $\mathfrak{p} = (2, 1+\sqrt{-5})$ is a non-principal ideal in $\mathcal{O}_K$.

Yang:

Example 2.3. Let $K = \mathbb{Q}(\sqrt{10})$ with ring of integers $\mathcal{O}_K = \mathbb{Z}[\sqrt{10}]$. The class number of K is 2, and the ideal $\mathfrak{p} = (2, \sqrt{10})$ is a non-principal ideal in $\mathcal{O}_K$.

Yang: To be continued.

Date: January 31, 2026, **Author**: Tianle Yang, My Homepage

Definition 2.4. Let $\overline{\mathcal{L}}$ be a hermitian line bundle on $\operatorname{Spec}\mathcal{O}_K$. The *degree* of $\overline{\mathcal{L}}$ is defined as

$$\widehat{\deg}\,\overline{\mathcal{L}} := \log \#(\mathcal{L}/\mathcal{O}_K s) - \sum_{\sigma: K\hookrightarrow\mathbb{C}} \log \|s\|_\sigma,$$

where s is any non-zero section of $\mathcal{L}$. Yang: To be revised

Definition 2.5. An *arithmetic divisor* on $\operatorname{Spec}\mathcal{O}_K$ is a pair (D, g), where

- D is a Weil divisor on $\operatorname{Spec}\mathcal{O}_K$, i.e., a formal sum of closed points with integer coefficients;
- $g = (g_\sigma)_{\sigma: K\hookrightarrow\mathbb{C}}$ is a collection of Green's functions $g_\sigma : (\operatorname{Spec}\mathcal{O}_K)_\sigma(\mathbb{C}) \setminus \{D\} \to \mathbb{R}$ for each embedding $\sigma : K \hookrightarrow \mathbb{C}$.

Yang: To be revised

Definition 2.6. Let (D, g) be an arithmetic divisor on $\operatorname{Spec}\mathcal{O}_K$. The *degree* of (D, g) is defined as

$$\widehat{\deg}(D, g) := \sum_{x\in\operatorname{Spec}\mathcal{O}_K} n_x \log N(x) + \sum_{\sigma: K\hookrightarrow\mathbb{C}} g_\sigma(x_\sigma),$$

where $D = \sum_x n_x x$ and $N(x)$ is the norm of the closed point x. Yang: To be revised

Proposition 2.7. There is a one-to-one correspondence between isomorphism classes of hermitian line bundles on $\operatorname{Spec}\mathcal{O}_K$ and arithmetic divisors on $\operatorname{Spec}\mathcal{O}_K$. Moreover, this correspondence preserves degrees, i.e., for a hermitian line bundle $\overline{\mathcal{L}}$ corresponding to an arithmetic divisor (D, g), we have

$$\widehat{\deg}\,\overline{\mathcal{L}} = \widehat{\deg}(D, g).$$

Yang: To be revised

Proposition 2.8. Let K be a number field with ring of integers $\mathcal{O}_K$. There is an exact sequence

$$0 \to \mu_K \to \mathcal{O}_K^\times \xrightarrow{\log_K} \mathbb{R}^r \xrightarrow{\iota} \widehat{\mathrm{Cl}}(\mathcal{O}_K) \to \mathrm{Cl}(\mathcal{O}_K) \to 0,$$

where μ_K is the group of roots of unity in K, $\widehat{\mathrm{Cl}}(\mathcal{O}_K)$ is the arithmetic class group of $\mathcal{O}_K$, and $\mathrm{Cl}(\mathcal{O}_K)$ is the usual class group of $\mathcal{O}_K$. Yang: To be revised

2.2 Geometry of numbers

Definition 2.9. Let $\overline{\mathcal{L}}$ be a hermitian line bundle over $\operatorname{Spec}\mathcal{O}_K$. The *successive minima* $\lambda_1(\overline{\mathcal{L}}), \ldots, \lambda_r(\overline{\mathcal{L}})$ are defined as

$$\lambda_i(\overline{\mathcal{L}}) := \inf\{\lambda > 0 : \dim_K K\{s \in H^0(\operatorname{Spec}\mathcal{O}_K, \mathcal{L}) : \|s\|_\sigma \le \lambda \text{ for all } \sigma\} \ge i\}.$$

Yang: To be revised

2.3 Riemann-Roch

Definition 2.10. Let $\overline{\mathcal{L}}$ be a hermitian line bundle over $\operatorname{Spec}\mathcal{O}_K$. The *Euler-Minkowski characteristic* $\chi(\overline{\mathcal{L}})$ is defined as

$$\chi(\overline{\mathcal{L}}) := \log \frac{\operatorname{vol}(B(\overline{\mathcal{L}}))}{\operatorname{vol}(\mathcal{L}_{\mathbb{R}}/\mathcal{L})},$$

where $B(\overline{\mathcal{L}}) = \{s \in \mathcal{L}_{\mathbb{R}} : \|s\|_\sigma \leq 1 \text{ for all } \sigma\}$ and the volumes are computed with respect to the Lebesgue measure. Yang: To be revised

3 Hermitian line bundles

Let K be a number field with ring of integers $\mathcal{O}_K$. Let $\mathcal{X}$ be an arithmetic variety over $\operatorname{Spec}\mathcal{O}_K$ of relative dimension n. Yang: flat,separated,of finite type,

For every $\sigma : \mathcal{O}_K \to \mathbb{C}$, we denote by $\mathcal{X}_\sigma(\mathbb{C}) := \mathcal{X} \times_{\operatorname{Spec}\mathcal{O}_K,\sigma} \operatorname{Spec}\mathbb{C}$ the complex analytic space associated to the base change of $\mathcal{X}$ along σ.

3.1 Definitions and examples

Definition 3.1. A *hermitian line bundle* $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma\in M_K^\infty})$ on $\mathcal{X}$ consists of

- a line bundle $\mathcal{L}$ on $\mathcal{X}$;
- for each embedding $\sigma : K \hookrightarrow \mathbb{C}$ (equivalently, each infinite place $\sigma \in M_K^\infty$), a smooth hermitian metric $\|\cdot\|_\sigma$ on the complex line bundle $\mathcal{L}_\sigma(\mathbb{C})$ over $\mathcal{X}_\sigma(\mathbb{C})$, which is invariant under the complex conjugation map.

Yang: To be checked.

Example 3.2. Consider $\mathcal{X} = \mathbb{P}^n_{\mathcal{O}_K}$ and $\mathcal{L} = \mathcal{O}_{\mathbb{P}^n_{\mathcal{O}_K}}(1)$. There is a class of natural hermitian metrics on $\mathcal{O}_{\mathbb{P}^n_{\mathcal{O}_K}}(1)_\sigma(\mathbb{C})$ called the *Fubini-Study metrics* constructed as follows.

Recall that $\mathbb{P}^n$ represents the functor that sends a scheme S to the set of isomorphism classes of pairs $(\mathcal{E}, \phi)$, where $\mathcal{E}$ is a rank $n+1$ vector bundle on S and $\phi : \mathcal{E} \twoheadrightarrow \mathcal{L}$ is a surjective morphism onto a line bundle $\mathcal{L}$ on S. And the universal pair on $\mathbb{P}^n$ is given by $\mathcal{O}_{\mathbb{P}^n}^{\oplus n+1} \twoheadrightarrow \mathcal{O}_{\mathbb{P}^n}(1)$.

For each $\sigma \in M_K^\infty$, fix a hermitian metric $\|\cdot\|$ on $\mathbb{C}^{n+1}$. Every point $x \in \mathbb{P}^n_{\mathcal{O}_K}(\sigma)$, it corresponds to a surjective morphism $\phi_x : \mathbb{C}^{n+1} \twoheadrightarrow L_x$ with L_x the fiber of $\mathcal{L}_\sigma(\mathbb{C})$ at x. Explicitly, every point $x = [z_0 : z_1 : \cdots : z_n] \in \mathbb{P}^n(\mathbb{C})$ corresponds to the surjective morphism

$$\phi_{[z_0:z_1:\cdots:z_n]} : \mathbb{C}^{n+1} \twoheadrightarrow \mathbb{C} \cong L_x, \quad (x_0, x_1, \dots, x_n) \mapsto x_0z_0 + x_1z_1 + \cdots + x_nz_n.$$

The fixed hermitian metric induces the quotient hermitian metric $\|\cdot\|_x$ on L_x. Hence we obtain a hermitian metric $\|\cdot\|_\sigma$ on $\mathcal{O}_{\mathbb{P}^n_{\mathcal{O}_K}}(1)_\sigma(\mathbb{C})$. This construction defines a hermitian line bundle $\overline{\mathcal{O}_{\mathbb{P}^n_{\mathcal{O}_K}}(1)}$.
Yang: To be checked.

Definition 3.3. Let $\overline{\mathcal{L}}_1$ and $\overline{\mathcal{L}}_2$ be hermitian line bundles on $\mathcal{X}$.

- The *tensor product* $\overline{\mathcal{L}}_1 \otimes \overline{\mathcal{L}}_2$ is defined as the line bundle $\mathcal{L}_1 \otimes \mathcal{L}_2$ on $\mathcal{X}$ equipped with the metrics $\|s_1 \otimes s_2\|_\sigma = \|s_1\|_{\sigma,1} \cdot \|s_2\|_{\sigma,2}$ on $(\mathcal{L}_1 \otimes \mathcal{L}_2)_\sigma(\mathbb{C})$
- The *dual* $\overline{\mathcal{L}}_1^\vee$ is defined as the dual line bundle $\mathcal{L}_1^\vee$ on $\mathcal{X}$ equipped with the dual metrics $\|\cdot\|_{\sigma,1}^\vee$ on $(\mathcal{L}_1^\vee)_\sigma(\mathbb{C})$ for each $\sigma \in M_K^\infty$.

Yang: To be checked.

Proposition 3.4. The set of isomorphism classes of hermitian line bundles on $\mathcal{X}$ forms an abelian group under the tensor product operation. The group is called the *arithmetic Picard group* of $\mathcal{X}$ and is denoted by $\widehat{\mathrm{Pic}}(\mathcal{X})$.

Proof. Direct verification. □

Note that for every $\sigma \in M_K^\infty$, any two smooth hermitian metrics on $\mathcal{L}_\sigma(\mathbb{C})$ differ by a smooth function on $\mathcal{X}_\sigma(\mathbb{C})$. Explicitly, for any two smooth hermitian metrics $\|\cdot\|_\sigma$ and $\|\cdot\|'_\sigma$ on $\mathcal{L}_\sigma(\mathbb{C})$, there exists a unique smooth function $f : \mathcal{X}_\sigma(\mathbb{C}) \to \mathbb{R}$ such that

$$\|s\|'_\sigma = \mathrm{e}^{-f}\|s\|_\sigma$$

for all local sections s of $\mathcal{L}_\sigma(\mathbb{C})$. Conversely, given any smooth hermitian metric $\|\cdot\|_\sigma$ on $\mathcal{L}_\sigma(\mathbb{C})$ and any smooth function $f : \mathcal{X}_\sigma(\mathbb{C}) \to \mathbb{R}$, we can define a new smooth hermitian metric $\|\cdot\|'_\sigma$ on $\mathcal{L}_\sigma(\mathbb{C})$ by the above formula.

Example 3.5. Let $\mathcal{X} = \mathbb{P}^n_{\mathbb{Z}}$. Then we have

$$\widehat{\mathrm{Pic}}(\mathbb{P}^n_{\mathbb{Z}}) \cong \mathbb{Z} \times \mathcal{C}^\infty(\mathbb{P}^n(\mathbb{C}), \mathbb{R}).$$

Here the generator of the first factor $\mathbb{Z}$ corresponds to the hermitian line bundle $\overline{\mathcal{O}_{\mathbb{P}^n_{\mathbb{Z}}}(1)}$ constructed in Example 3.2.

Definition 3.6. Let $f : \mathcal{Y} \to \mathcal{X}$ be a morphism of arithmetic varieties over $\operatorname{Spec} \mathcal{O}_K$. Let $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma \in M_K^\infty})$ be a hermitian line bundle on $\mathcal{X}$. The *pullback* $f^*\overline{\mathcal{L}}$ is defined as the line bundle $f^*\mathcal{L}$ on $\mathcal{Y}$ equipped with the metrics $\|s\|'_\sigma = \|f_\sigma^* s\|_\sigma$ on $(f^*\mathcal{L})_\sigma(\mathbb{C})$ for each $\sigma \in M_K^\infty$, where $f_\sigma : \mathcal{Y}_\sigma(\mathbb{C}) \to \mathcal{X}_\sigma(\mathbb{C})$ is the induced morphism of complex manifolds. Yang: To be checked.

Definition 3.7. Let $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma \in M_K^\infty})$ be a hermitian line bundle on $\mathcal{X}$. The set of *small sections* of $\overline{\mathcal{L}}$ is defined as

$$H^0(\mathcal{X}, \overline{\mathcal{L}}) := \{s \in H^0(\mathcal{X}, \mathcal{L}) : \|s(x)\|_\sigma \le 1,\ \forall \sigma \in M_K^\infty, x \in \mathcal{X}_\sigma(\mathbb{C})\}.$$

Yang: To be checked.

Example 3.8. Let $\mathcal{X} = \mathbb{P}^n_{\mathbb{Z}}$ and $\overline{\mathcal{L}} = \overline{\mathcal{O}_{\mathbb{P}^n_{\mathbb{Z}}}(1)}$ be the hermitian line bundle constructed in Example 3.2. We have

$$H^0(\mathbb{P}^n_{\mathbb{Z}}, \mathcal{L}) = \mathbb{Z}X_0 \oplus \mathbb{Z}X_1 \oplus \cdots \oplus \mathbb{Z}X_n,$$

where $X_0, X_1, \ldots, X_n$ are the standard homogeneous coordinates on $\mathbb{P}^n_{\mathbb{Z}}$. For every $s = \sum a_n X_n \in$

$H^0(\mathbb{P}^n_{\mathbb{Z}}, \mathcal{L})$ and $x = [x_0 : \cdots : x_n] \in \mathbb{P}^n(\mathbb{C})$, we have

$$\|s(x)\| = \inf_{x_0y_0+\cdots+x_ny_n=0} \left(\sum |a_k + y_k|^2\right)^{1/2} = \frac{|\sum a_k x_k|}{\|x\|},$$

where $\|x\| = (|x_0|^2 + |x_1|^2 + \cdots + |x_n|^2)^{1/2}$. When x varies, by Cauchy-Schwarz inequality, we have

$$\sup_{x\in\mathbb{P}^n(\mathbb{C})} \|s(x)\| = \sup_{x\in\mathbb{P}^n(\mathbb{C})} \frac{|\sum a_k x_k|}{\|x\|} = \|a\|.$$

Note that $a_n \in \mathbb{Z}$ for all n. Therefore, we have

$$H^0(\mathbb{P}^n_{\mathbb{Z}}, \overline{\mathcal{O}_{\mathbb{P}^n_{\mathbb{Z}}}(1)}) = \{0, \pm X_0, \pm X_1, \ldots, \pm X_n\}$$

and

$$\# H^0(\mathbb{P}^n_{\mathbb{Z}}, \overline{\mathcal{O}_{\mathbb{P}^n_{\mathbb{Z}}}(1)}) = 2n + 3.$$

Proposition 3.9. The set $H^0(\mathcal{X}, \overline{\mathcal{L}})$ is finite.

Definition 3.10. Let $\overline{\mathcal{L}}$ be a hermitian line bundle on $\mathcal{X}$. We set

$$h^0(\mathcal{X}, \overline{\mathcal{L}}) := \log \# H^0(\mathcal{X}, \overline{\mathcal{L}}).$$

The *volume* of $\overline{\mathcal{L}}$ is defined as

$$\mathrm{vol}(\overline{\mathcal{L}}) := \limsup_{m\to\infty} \frac{h^0(\mathcal{X}, \overline{\mathcal{L}}^{\otimes m})}{m^{n+1}/(n+1)!}.$$

Example 3.11. We continue with Example 3.8. For every integer $m \geq 1$, we have

$$H^0(\mathbb{P}^n_{\mathbb{Z}}, \mathcal{O}_{\mathbb{P}^n_{\mathbb{Z}}}(m)) = \bigoplus_{|\alpha|=m} \mathbb{Z}X^\alpha.$$

Let $t = \sum_{|\alpha|=m} a_\alpha X^\alpha \in H^0(\mathbb{P}^n_{\mathbb{Z}}, \mathcal{O}_{\mathbb{P}^n_{\mathbb{Z}}}(m))$ be a global section. Denote by M (resp. L) the geometric vector bundle associated to $\mathcal{O}_{\mathbb{P}^n_{\mathbb{Z}}}(m)$ (resp. $\mathcal{O}_{\mathbb{P}^n_{\mathbb{Z}}}(1)$).

We first focus on the affine chart $U_0 = \{X_0 \neq 0\}$ and let $s = X_0 \in H^0(\mathbb{P}^n, \mathcal{O}(1))$. For every point $x = [1 : x_1 : \cdots : x_n] \in U_0(\mathbb{C})$,

$$\|s(x)\| = \frac{1}{(1 + |x_1|^2 + \cdots + |x_n|^2)^{1/2}} \implies \|t(x)\| = \frac{|\sum a_\alpha x^\alpha|}{(1 + |x_1|^2 + \cdots + |x_n|^2)^{m/2}}.$$

Hence one can see that for any point $x = [x_0 : \cdots : x_n] \in \mathbb{P}^n(\mathbb{C})$, we have

$$\|t(x)\| = \frac{|\sum a_\alpha x^\alpha|}{\|x\|^m}$$

with $\|x\| = (|x_0|^2 + |x_1|^2 + \cdots + |x_n|^2)^{1/2}$.

Yang: In general, it seems difficult to compute $h^0(\mathbb{P}^n_{\mathbb{Z}}, \overline{\mathcal{O}(m)})$ explicitly. For example, when $m = 2$, we have that $\sup_{x\in\mathbb{P}^n(\mathbb{C})} \|t(x)\|$ is equal to the largest norm of eigenvalues of the matrix associated to the quadratic form (a_{ij}). Hence we just compute its asymptotic behavior when $m \to \infty$.

Note that we have

$$\frac{|\sum x^{\alpha}|}{\|x\|^m} \cdot \min_{|\alpha|=m} |a_{\alpha}| \le \|t(x)\| \le \frac{|\sum x^{\alpha}|}{\|x\|^m} \cdot \max_{|\alpha|=m} |a_{\alpha}|.$$

By the lore of inequalities (like Karamata's inequality), one can show that

$$\sup_{x\in\mathbb{P}^n(\mathbb{C})} \frac{|\sum x^{\alpha}|}{\|x\|^m} = \binom{m+n}{n}\frac{1}{(n+1)^{m/2}}.$$

Yang: To be checked.

3.2 Arithmetic intersection theory

4 Arithmetic Hilbert-Samuel theorem

Theorem 4.1. Let $\mathcal{X}$ be an arithmetic variety of absolute dimension d over $\operatorname{Spec} \mathcal{O}_K$, and let $\overline{\mathcal{L}}$ be a hermitian line bundle on $\mathcal{X}$. Assume that $\overline{\mathcal{L}}$ is nef. Then, we have

$$\operatorname{vol}(\overline{\mathcal{L}}) = \widehat{\deg}\left(\hat{c}_1(\overline{\mathcal{L}})^d\right).$$

5 Arithmetic fundamental inequality